

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 RENEWCOILS ENGINEER & SUPPLY CO.,LTD. Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnaury W.		
Certified By	09-Jun-2012	Mr.Amnaury W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnaury W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-012-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
SMPC/RNC-WS-002/10**



2004
Deming Application
Prize Winner

ISO 9001 QMS 0207/9520
ISO 14001 EMS 01005/043
TIS 10001 QMS 0201/0630
CHSAS 10001 CHSAS 02002/0602

Sheet 1 of 2

WELDING PROCEDURE SPECIFICATIONS (WPS)

Company Name SIAM MITSUI PTA CO., LTD. /
RENEW COILS ENGINEERING & SUPPLY CO., LTD.
Welding Procedure Specification no. SMPC/RNC-WS-002/10
Revision no. 0 Date August 16, 2010
Welding Process(es) 1/ GTAW 2/ SMAW
Impact Test Requirement ☐ Yes ☒ No

By Mr. Tossaporn Rojvattanakarn (Sign/Date)
Date August 16, 2010
Supporting PQR no.(s) SMPC/RNC-PQ-002/10
Type(s) 1/ Manual
2/ Manual

JOINTS (QW-402)

Joint Design Groove design used
Backing Without backing or with backing (if required)
Backing Material (Type) 1/ All backing material
(Refer to both backing and retainers) 2/ All backing material
☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other (Weld Metal)
Applicable of Tack Welding ☒ Yes ☐ No
Applicable of Repair Welding ☒ Yes ☐ No

Details

Groove design as per production drawing.

BASE METALS (QW-403)

P-No. 8 Group no. All* to P-no. 8 Group no. All*
OR Specification Type and Grade SA 312 TP304, 304L to SA 312 TP304, 304L
OR Chem. Analysis and Mech. Prop. Not applicable to Not applicable
Base Metal Thk. Range (T, mm.) Groove 5.0 - 36.52 mm. Fillet All thicknesses
Pipe Dia. Range Groove All diameters Fillet All diameter
Maximum Pass Thickness $\leq 1/2$ in. (13.0 mm.) ☒ Yes ☐ No
Other *Any group number for non-impact tested applications.

FILLER METALS (QW-404)

Spec. no. (SFA)	1/ <u>5.9</u>	2/ <u>5.4</u>
AWS no. (Class)	1/ <u>ER 308, 308L</u>	2/ <u>E308L-16</u>
F-no.	1/ <u>6</u>	2/ <u>5</u>
A-no.	1/ <u>8</u>	2/ <u>8</u>
Size of Filler Metals	1/ <u>Ø 2.0 or 2.4 mm.</u>	2/ <u>Ø 2.6, 3.2 or 4.0 mm</u>
Deposited Weld Metal Thk. Range (t, mm.)		
	Groove 1/ <u>12.52 mm (Max)</u>	2/ <u>24.0 mm (Max)</u>
	Fillet 1/ <u>All thicknesses</u>	2/ <u>All thicknesses</u>
Filler metal product form	1/ <u>Solid wire</u>	2/ <u>Not applicable</u>
Supplemental Filler Metal	1/ <u>Not applicable</u>	2/ <u>Not applicable</u>
Electrode-Flux (class)	1/ <u>Not applicable</u>	2/ <u>Not applicable</u>
Flux Type	1/ <u>Not applicable</u>	2/ <u>Not applicable</u>
Flux Trade Name	1/ <u>Not applicable</u>	2/ <u>Not applicable</u>
Consumable Insert	1/ <u>None</u>	2/ <u>None</u>
Other	<u>None</u>	

FORM : WPS SEC.IX R.3 Page 1 of 2, 25 Aug. 2010

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บริษัท สยามมิทซุย พัททิง จำกัด
1 ถนนปิ่นเกล้าพิเศษ แขวง ปิ่นเกล้า เขต ปิ่นเกล้า กรุงเทพมหานคร 10800
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1 Siam Cement Rd., Bangsue,
Bangkok 10800, Thailand
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8 ถนนปิ่นเกล้าพิเศษ แขวง ปิ่นเกล้า เขต ปิ่นเกล้า กรุงเทพมหานคร 21150
เลขที่ 5-2 อาคาร 5 ชั้น 500 จ.ระยอง 21150
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Muang District, Rayong Province 21150, Thailand
Tel : 66 3868 5100 Fax : 66 3868 5099



2004
Daring Application
Prize Winner



ISO 9001 OHS 02075620
ISO 14001 OHS 01055243
TS 16949 OHS 02075620
OHSAS 18001 OHSAS 02075620

Sheet 2 of 2

WPS no. SMPC/RNC-WS-002/10 Rev. 0

POSITIONS (QW-405)

Position of Groove 1/ All 2/ All
Welding Progression : Up hill Down -
Position(s) of Fillet 1/ All 2/ All

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range (NO PWHT applied) °C
Time Range - hr.
Other None

PREHEAT (QW-406)

Preheat Temp. Min. 10 °C
Interpass Temp. Max. 150 °C
Preheat Maintenance Cool in still air
Other None

GAS (QW-408)

Percent Composition			
	Gas(es)	Mixture	Flow Rate
Shielding	<u>Argon</u>	<u>Welding Grade</u>	<u>10-20 l/min</u>
Trailing	<u>None</u>	<u>-</u>	<u>-</u>
Backing	<u>Argon²⁾</u>	<u>Welding Grade</u>	<u>10-25 l/min</u>

ELECTRICAL CHARACTERISTICS (QW-409)

Current AC or DC 1/ DC 2/ AC or DC Polarity 1/ EN 2/ EP for DC
Amps (Range) See table below Volts (Range) See table below
Tungsten Electrode Size and Type Ø 2.4 mm / 2% Thoriated (EWTh-2, AWS A5.12) for GTAW process
Mode of Metal Transfer for GMAW/FCAW Not applicable
Electrode Wire Feed Speed Range Not applicable
Heat input (Range) Not Specified.

TECHNIQUE (QW-410)

String or Weave Bead 1/ Both 2/ Both
Orifice or Gas Cup Size 1/ No. 6 - 10 2/ NA
Initial and Interpass Cleaning Power brushing and / or Grinding
Method of Back Gouging Any mechanical method
Oscillation None
Contact Tube to Work Distance 1/ Not applicable 2/ Not applicable
Multiple or Single Pass (per side) 1/ Multiple 2/ Multiple
Multiple or Single Electrodes 1/ Single 2/ Single
Travel Speed (Range) See table below
Peening None
Other Out of chamber conventional torch welding technique to be employed.

Weld Layer(s)	Process(es)	Filler Metal		Current		Volt Range	Travel Speed Range: cm/minute	Gas Flow Rate (Ar.)	
		Class	Dia.	Type Polar.	Amp. Range			Shielding	Backing
Root	GTAW	ER 308L	Ø 2.0 mm.	DC/EN	100-130 A	11-16 V	4 - 10	10-20 l/min	10-25 l/min.
	GTAW	ER 308L	Ø 2.4 mm.	DC/EN	120-165 A	11-17 V	4 - 10	10-20 l/min	10-25 l/min.
Fill & Over	SMAW	E308L-16	Ø 2.6 mm.	AC or DC/EP	80-110 A	18-26 V	5 - 14	-	-
	SMAW	E308L-16	Ø 3.2, 4.0 mm.	AC or DC/EP	90-140 A	19-28 V	5 - 14	-	-

Note:

- 1) This WPS shall be applied to weld joints where impact test is not required.
- 2) Gas backing shall only be applied to single-welded groove joint to be penetrated fully.
- 3) The maximum weave width should not exceed three(3) times the core diameter of filler metal to be used.

FORM : WPS SBC/IX R.3 Page 1 of 2, 25 Aug. 2010

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ซอย 5-2 ต.ปิ่นเกล้า อ.ปิ่นทอง จ.นนทบุรี 21150
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PROCEDURE QUALIFICATION RECORD (PQR)

Company Name SIAM MITSUI PTA CO., LTD. / RENEW COILS ENGINEERING & SUPPLY CO., LTD.Procedure Qualification Record no. SMPC/RNC-PQ-002/10 Date August 16, 2010WPS no. SMPC/RNC-WS-002/10 Revision no. 0Welding Process(es) 1/ GTAW 2/ SMAWTypes (Manual, Automatic, Semi-Automatic) 1/ Manual 2/ ManualImpact Test Requirement ☐ Yes ☒ No

JOINTS (QW-402)

SEE PAGE 3 OF 3

BASE METALS (QW-403)

Material Spec. SA 312 to SA 312
Type/Grade or UNS Number TP 304L to TP 304L
P-No. 8 Group No. 1 to P-No. 8 Group No. 1
Thickness of Test Coupon, T 18.26 mm.
Diameter of Test Coupon NPS 8"
Maximum Pass Thickness 4.0 for SMAW mm.
Other No t pass > 1/2"(13.0 mm.) for SMAW.

FILLER METALS (QW-404)

SFA Specification 1/ 5.9 2/ 5.4
AWS Classification 1/ ER 308L 2/ E308L-16
Filler Metal F.no. 1/ 6 2/ 5
Weld Metal Analysis A. No. 1/ 8 2/ 8
Size of Filler Metal 1/ Ø 2.4 mm. 2/ Ø 2.6, 3.2 mm.
Filler Metal Product Form 1/ Solid wire 2/ Not applicable
Supplemental Filler Metal 1/ Not applicable 2/ Not applicable
Electrode Flux Classification 1/ Not applicable 2/ Not applicable
Flux Type 1/ Not applicable 2/ Not applicable
Flux Trade Name 1/ Not applicable 2/ Not applicable
Weld Metal Thickness 1/ 6.26 mm. 2/ 12.00 mm.
Other Brand Name : KOBELCO TG-S308L for GTAW, KOBE NC-38 for SMAW

POSITIONS (QW-405)

Position of Groove 1/ 6G 2/ 6G
Welding Progression (Uphill, Downhill)
1/ Uphill 2/ Uphill
Other None

PREHEAT (QW-406)

Preheat Temp. Ambient (~ 30) °C
Interpass Temp. 150 (Max) °C
Other Cool in still air

POSTWELD HEAT TREATMENT (QW-407)

Temperature (NO PWHT applied) °C
Time - hr.
Other None

GAS (QW-408)

	Gas(es)	Mixture	Flow Rate
Shielding	<u>Argon</u>	<u>99.95%</u>	<u>14 l/min</u>
Trailing	<u>None</u>	<u>-</u>	<u>-</u>
Backing	<u>Argon</u>	<u>99.95%</u>	<u>20 l/min</u>
Other	<u>None</u>	<u>-</u>	<u>-</u>

ELECTRICAL CHARACTERISTICS (QW-409)

Current 1/ DC 2/ DC
Polarity 1/ EN 2/ EP
Amps. 1/ See sheet 3,4 2/ See sheet 3,4
Volts 1/ See sheet 3,4 2/ See sheet 3,4
Tungsten Electrode Size Ø 2.4 mm./2% Thoriated
Mode of Metal Transfer for GMAW (FCAW)
None
Other None

TECHNIQUE (QW-410)

Travel Speed 1/ See sheet 3,4 2/ See sheet 3,4
String or Weave Bead 1/ both 2/ both
Oscillation NA
Multipass or Single Pass (per side) Multipass
Single or Multiple Electrodes Single
Other None

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2004
Daring Application
Prize Winner



ISO 9001 OHS 22075/250
ISO 14001 EMS 01005043
TIS 18001 OHS 02018/088
OHSAS 18001 OHSAS 02002002

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PQR Back

PQR no. SMPC/RNC-PQ-002/10

TENSILE TEST (QW-150)

Specimen no.	Width (mm.)	Thickness (mm.)	Area (mm ²)	Ultimate Total Load (N)	Ultimate Unit Stress (N/mm ²)	Type of Failure and Location
TS 302/2010	19.46	18.06	351.45	250,000	583.30	Failure out of weld
TS 303/2010	19.38	18.58	360.08	208,000	577.65	Failure out of weld

GUIDED BEND TEST (QW-160)

Specimen no.	Type & Figure No. of Bend Tests	Results	Remarks
SB 345/2010	Transverse Side Bend (QW 462.2)	Acceptable	No Discontinuity
SB 346/2010	Transverse Side Bend (QW 462.2)	Acceptable	No Discontinuity
SB 347/2010	Transverse Side Bend (QW 462.2)	Acceptable	No Discontinuity
SB 348/2010	Transverse Side Bend (QW 462.2)	Acceptable	No Discontinuity

TOUGHNESS TESTS (QW-170)

Specimen no.	Notch Location	Specimen Size	Test Temp. °C	Impact Values			Drop Weight Break	
				ft.-lb or J	%Shear	Mils	Break	No Break
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-

FILLET-WELD TEST (QW-180)

Results - Satisfactory : ☐ Yes ☐ No Penetration into Parent Metal : ☐ Yes ☐ No

Macro - Results : NA

OTHER TESTS

Type of Test RADIOGRAPHIC TEST AS PER REPORT No. R10-4248-I-P42

Deposit Analysis NA

Other None

Welder's name Mr. Boontham Jandong Clock no. - Stamp no. W-04

Test Conducted by SIWA TESTING INSPECTION & CONSULTING CO., LTD. Laboratory Test no. R10-4248-2,3-P42

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Boiler and Pressure Vessel Code.

Manufacturer SIAM MITSUI PTA CO., LTD.

Date Certify _____

FORM : PQR SECIX R.3 Page 2 of 3, 19 Aug. 2010

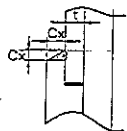
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บริษัท สยามมิทซุย พาร์ค จำกัด
1 ถนนสุขุมวิทซอย 10 แขวง 10800
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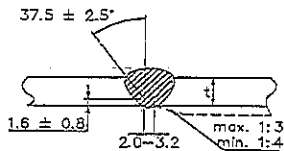
บริษัท สยามมิทซุย พาร์ค จำกัด
8 ถนนพหลโยธินซอย 1 แขวง 21150
ซอย 2-2 แขวง 21150 โทรสาร : 0 3868 5100 โทรสาร : 0 3868 5099
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ATTACHED FIGURE JOINT SKETCH



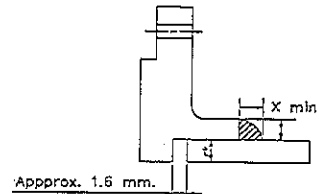
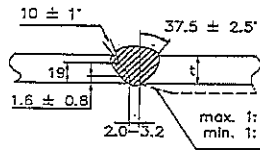
C_x (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



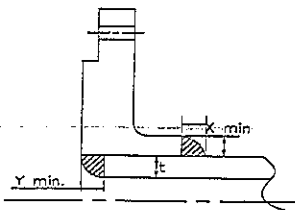
Misalignment (max.) = 1.5 mm.

JOINT No. B1 & B2



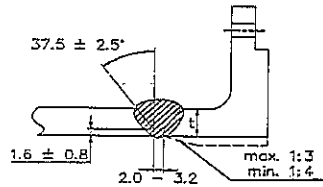
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



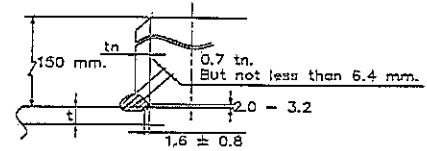
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



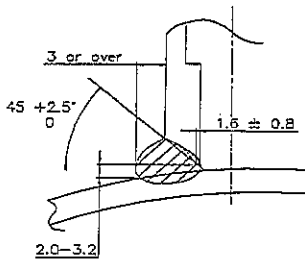
Misalignment (max.) = 1.5 mm.

JOINT No. B3

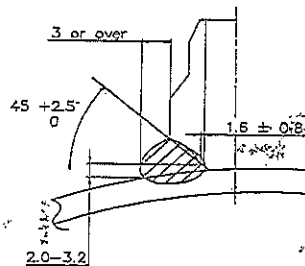


Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

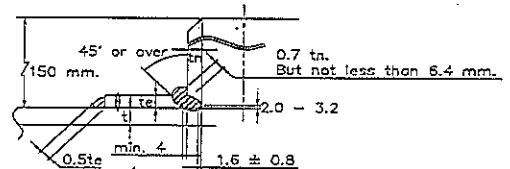
JOINT No. B4



JOINT No. C1

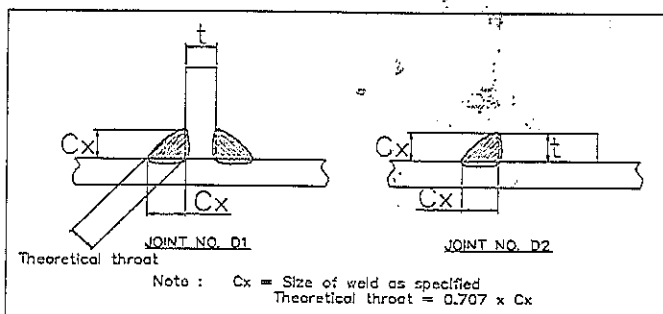


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

JOINT No. B5





บริษัท สิก- เทสติ้ง อินสเปคชั่น แอนด์ คอนซัลติง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO.,LTD.

3 of 4

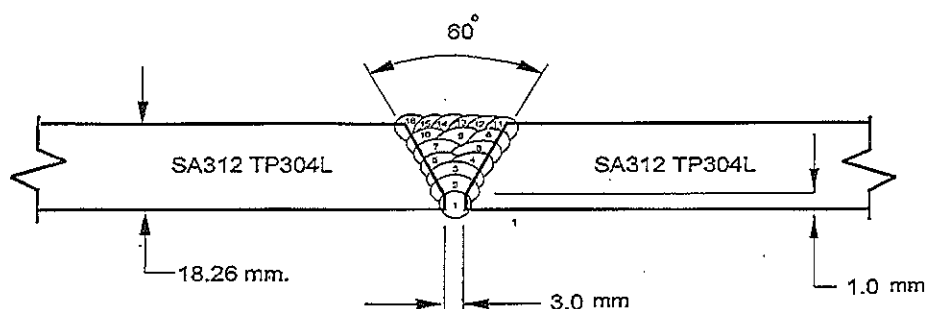
Actual Records of Welding Variable

ACC. TO ASME SECTION IX WELDING VARIABLE

WPS NO. SMPC/RNC-WS-002/10 Revision NO. 0

WELDER NO. W-04

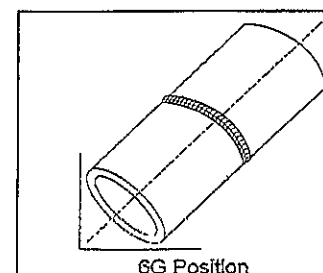
Front side



Back side

Type of material ☒ Pipe (NPS 8" SCH 120) ☐ Plate
Backing ☒ Yes for SMAW ☒ No for GTAW
Retainers ☐ Yes ☒ No

PASS SEQUENCE



Side	Pass	Process	Electrode or Filler		Amps./ Polarity	Volts	Travel Speed (cm./minute)	Backing gas flow rate	Shielding gas flow rate
			Type	Size					
Front	1	GTAW	ER 308L	Ø 2.4 mm.	112-114 DC/EN	13 - 14 V.	4.62	20 l/min.	14 l/min
	2	GTAW	ER 308L	Ø 2.4 mm.	145-148 DC/EN	14 - 15 V.	9.64	20 l/min.	14 l/min
	3	GTAW	ER 308L	Ø 2.4 mm.	155-159 DC/EN	14 - 15 V.	8.76	20 l/min.	14 l/min
	4	SMAW	E308L-16	Ø 3.2 mm.	99-102 DC/EP	21 - 22 V.	11.81	-	-
	5	SMAW	E308L-16	Ø 3.2 mm.	103-108 DC/EP	23 - 24 V.	10.88	-	-
	6	SMAW	E308L-16	Ø 3.2 mm.	102-105 DC/EP	23 - 24 V.	10.49	-	-
	7	SMAW	E308L-16	Ø 3.2 mm.	102-105 DC/EP	23 - 24 V.	6.49	-	-
	8	SMAW	E308L-16	Ø 3.2 mm.	102-105 DC/EP	23 - 24 V.	7.86	-	-
	9	SMAW	E308L-16	Ø 3.2 mm.	102-105 DC/EP	23 - 24 V.	6.45	-	-
	10	SMAW	E308L-16	Ø 3.2 mm.	102-105 DC/EP	23 - 24 V.	6.69	-	-
See continuous next page.									

Witnessed by :

J. Patiphan
Mr. Patiphan J.

Date :

AUGUST 4, 2010



บริษัท ซีว: เทสติ้ง อินสเปคชั่น แอนด์ คอนซัลติง จำกัด
SIWA TESTING INSPECTION & CONSULTING CO.,LTD.

Sheet 4 of 4

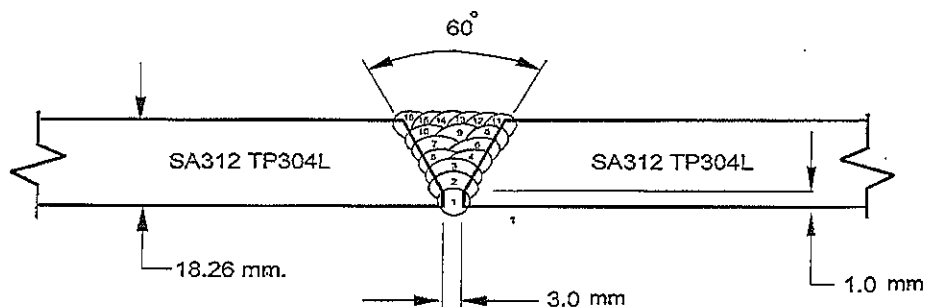
Actual Records of Welding Variable

ACC. TO ASME SECTION IX WELDING VARIABLE

WPS NO. SMPC/RNC-WS-002/10 Revision NO. 0

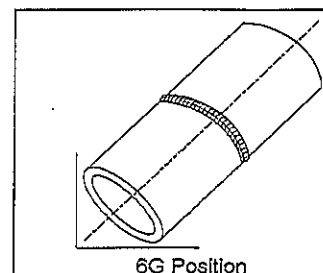
WELDER NO. W-04

Front side



Back side

Type of material ☒ Pipe (NPS 8" SCH 120) ☐ Plate
Backing ☒ Yes for SMAW ☒ No for GTAW
Retainers ☐ Yes ☒ No



PASS SEQUENCE

Side	Pass	Process	Electrode or Filler		Amps./ Polarity	Volts	Travel Speed (cm./minute)	Backing gas flow rate	Shielding gas flow rate
			Type	Size					
Front	11	SMAW	E308L-16	Ø 2.6 mm.	83-88 DC/EP	21 - 22 V.	11.51	-	-
	12	SMAW	E308L-16	Ø 2.6 mm.	83-88 DC/EP	21 - 22 V.	11.08	-	-
	13	SMAW	E308L-16	Ø 2.6 mm.	83-88 DC/EP	21 - 22 V.	10.47	-	-
	14	SMAW	E308L-16	Ø 2.6 mm.	83-88 DC/EP	21 - 22 V.	10.18	-	-
	15	SMAW	E308L-16	Ø 2.6 mm.	83-88 DC/EP	21 - 22 V.	10.11	-	-
	16	SMAW	E308L-16	Ø 2.6 mm.	83-88 DC/EP	21 - 22 V.	10.34	-	-

Witnessed by :

J. Patiphan
Mr. Patiphan

Date :

AUGUST 4, 2010

[illegible]



บริษัท ซีวี: เอสดีบี อีเอสพีเคเอ็ม จำกัด
SIWA TESTING INSPECTION & CONSULTING CO., LTD.

Welder Cert. no. 256/2010

WELDER PERFORMANCE QUALIFICATIONS (WPQ)

Company's name : SIAM MITSUI PTA CO., LTD.

Welder's name : Mr. Boontham Jandong

Identification no. 3 2512 00585 69 3



Test Description

Identification of WPS followed or joint type Butt joint (Single V-groove) ☒ Test coupon ☐ Production Weld
Specification and type/grade or UNS Number of Base Metal(s) SA 312 TP 304L Thickness 18.26 mm.
Filler metal or electrode specification (SFA) 5.9 + 5.4 Classification(s) ER 308L + E308L-16

Testing Variables and Qualification Limits

Welding Variables (QW-350)

Welding process(es) used

Type (ie; manual, semi-automatic) used

Backing (with/without)

Base metal (P-Number) 8 Group 1 to (P-Number)☐ Plate ☒ Pipe (enter diameter, if pipe)Groove
Fillet

Filler metal F-Number(s)

Filler metal product form (solid/cored/flux-cored) GTAW/PAW

Consumable insert for GTAW or PAW

Weld deposit thickness for each welding process

Process 1: GTAW 3 layers minimum ☒ Yes ☐ NoProcess 2: SMAW 3 layers minimum ☒ Yes ☐ No

Welding position

Groove weld test

Plate & Pipe O.D. > 24" (610 mm)

Pipe O.D. ≤ 24" (610 mm)

Fillet weld test

Vertical Progression (uphill or downhill)

Backing gas for GTAW, PAW or GMAW/FCAW

GMAW/FCAW transfer mode

GTAW welding current type/polarity

Actual Values

GTAW + SMAW

Manual

Without backing for GTAW

With backing for SMAW

8 Group 1

NPS 8" SCH. 120

Not applicable

6 + 5

Solid wire filler metal

None

6.26 mm.

12.00 mm.

6G

Not applicable

Uphill

None

Not applicable

DC/EN

Range Qualified

GTAW and/or SMAW

Manual

With or without backing

With backing or back gouging

P-No.1 through P-No.15E, P-No.34, and
P-No.41 through P-No.49 metals of similar
chemical composition to these metals.

OD 2 7/8" to Unlimited

All thicknesses & diameters

With or without backing : 6

With backing : 5,1

With solid filler metal (GTAW)

With or without consumable insert

12.5 mm.(Max.) & all fillet sizes

24.0 mm.(Max.) & all fillet sizes

All positions

All positions

All positions

Uphill

With or Without backing gas

-

DC/EN

RESULTS

Visual examination of Completed Weld (QW-302.4)

ACCEPTABLE

Guided Bend Test (QW-160)

☐ Transverse root and face [QW-462.3(a)];☐ Longitudinal root and face [QW-462.3(b)];☐ Side [QW-462.2)]☐ Pipe bend specimen, corrosion-resistant [QW-462.5(c)]☐ Plate bend specimen, corrosion-resistant [QW-462.5(d)]

Type and Specimen no.	Results
None	-
None	-

Type and Specimen no.	Results
None	-
None	-

Radiographic Examination (QW-191, For alternative qualification of groove welds by radiography)

Radiographic examination result (QW-191.2)

ACCEPTABLE

Test conducted and evaluated by:

SIWA TESTING INSPECTION & CONSULTING CO., LTD.

Report no.

R10-4248-1-P42

Fillet Weld Test (See QW-180)

Fracture test (QW-181.2)

Not applicable

Length and percent of defects

Not applicable

☐ Fillet weld in plate [QW-462.4(b)]☐ Fillet weld in pipe [QW-462.4(c)]

Macro Examination (QW-184) Not applicable Fillet leg size (in.)

X

Concavity/convexity (in.)

-

Other Tests

Type of test

Mechanical tests conducted and evaluated by:

Report no.

Welding Test Witnessed / Reviewed by:

J. Patiphan J.

K. Puchana K.

Witnessed date:

AUGUST 4, 2010

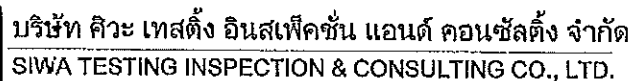
SIWA TESTING INSPECTION & CONSULTING CO., LTD.

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of the Section IX of the ASME BOILER AND PRESSURE VESSEL CODE.

Certified by:

Date:

(Manufacturer's representative)



Report no. : R10-4248-2-P.42 Page 1 of 1

Test Date: August 11, 2010

Client : Siam Mitsui PTA Co., Ltd.

Tested Product : Test Coupon for PQR No. SMPC/RNC-PQ-002/10 (Welded Pipe Ø 8")

Material : SA312 TP304L

Project : ~

Applicable Standard : ASME Sec. IX : 2007

Thickness : 18.26 mm

No.	Specimen No.	Size of Specimen (mm)	Gauge Length (mm)	Elongation (mm)	Elongation (%)	Tested Area (mm ²)	Load		Stress		Remark
							Yield (N)	Ultimate (N)	Yield (N/mm ²)	Ultimate (N/mm ²)	
1	TS 302 / 2010	19.46 x 18.06	-	-	-	351.45	-	205,000	-	583.30	FOOW - Accept
2	TS 303 / 2010	19.38 x 18.58	-	-	-	360.08	-	208,000	-	577.65	FOOW - Accept

Remark : FAW - Failure at weld

FOOW - Failure out of weld

Total	2	Specimen(s)	☐ Attached Report =	Page(s)
Operated by	Nopporn Karnjanajutaphan 		Reviewed and Supervised by :  (Patoomporn Phongchotip)	
Certified by	 (Patoomporn Phongchotip / Project Engineer)		Date : 16 AUG 2010	

FORM : Tensile Test R.13FuK17

The present inspection has been carried out to the best of our knowledge and belief. By signing this inspection report, neither the inspector nor the company and its representatives shall be liable in any manner for any personal injury, properties damage or loss of any kind arising from or concerned with this inspection.

สำนักงานกรุงเทพ 195 ซอยเพชรเกษม 65 ถนนเพชรเกษม แขวงหลักสอง เขตบางเขน กรุงเทพฯ 10160
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Tel. (66)(0) 2444-3845 (15 lines) Fax. (66)(0) 2444-3028, 2444-3027 Email. info@stwater.co.th

สำนักงานระยอง 267/306-307 ถนนสุขุมวิท ต.มาบตาพุด อ.เมือง จ.ระยอง 21150
Rayong Office 267/306-307 Sukhumvit Rd. Map Ta Phut, Amphur Muang, Rayong 21150, Thailand
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ISSUE NO. 1

ORIGINAL

Specimen No.	Type of Bending	Size of Specimen	Fracture Location	Remark
SB 345 / 2010	Transverse Side Bend	9.5 x 18.26	No Discontinuity	Accept
SB 346 / 2010	Transverse Side Bend	9.5 x 18.26	No Discontinuity	Accept
SB 347 / 2010	Transverse Side Bend	9.5 x 18.26	No Discontinuity	Accept
SB 348 / 2010	Transverse Side Bend	9.5 x 18.26	No Discontinuity	Accept

[illegible]

Total :		4	Specimen(s)	☐ Attached Report =	Page(s)
Operated by	Nopporn Karnjanajutaphan		Reviewed and Supervised by :  (Patoomporn Phongchotip)		
Certified by	  (Patoomporn Phongchotip / Project Engineer)		Date : 16 AUG 2010		